City of
Maple Grove

CITY PROJECT 87-26
CROWN POINTE

CONSTRUCTION PLANS FOR
UTILITY AND STREET IMPROVEMENTS

SHEET INDEX

1	TITLE SHEET
2	PROJECT INFORMATION & TYPICAL SECTION
3-5	SANITARY SEWER & WATERMAIN PLAN & PROFILE
6-8	STREET & STORM SEWER PLAN & PROFILE

PREPARED BY
bennett ringrose wolsfeld jarvis gardner inc.
THRESHER SQUARE 700 THIRD ST. S., MINNEAPOLIS, MN 55415

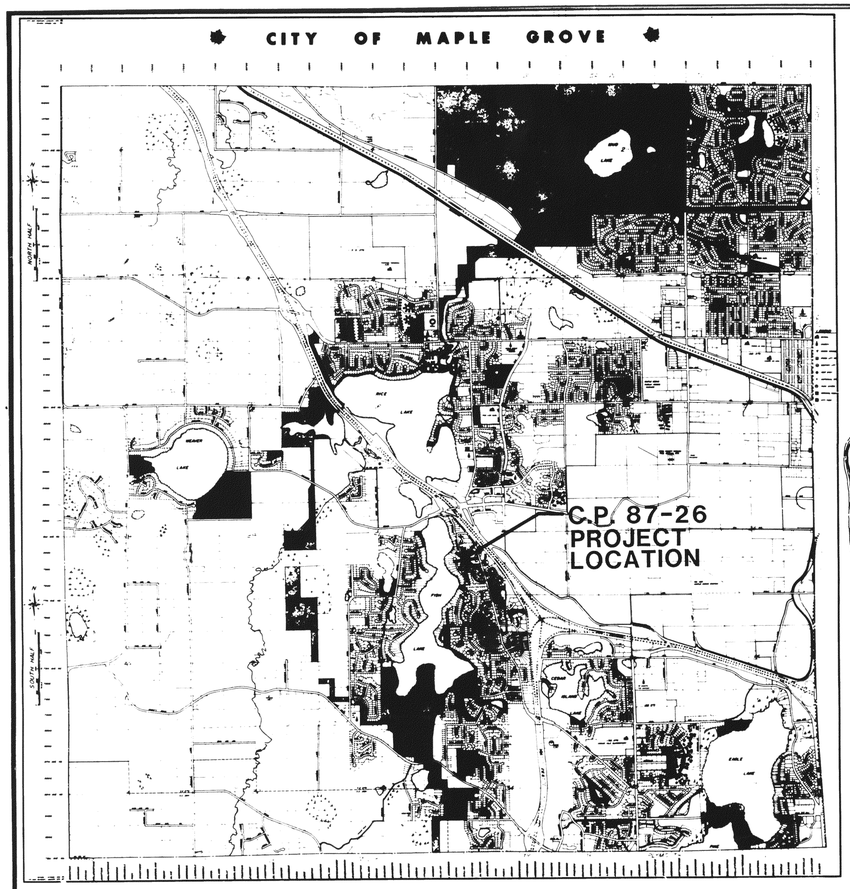
BENCH MARKS
1 - TOP NUT HYD. BETWEEN LOTS 7 BLK. 1, STEPNEY RIDGE & LOT 1 BLK 1, CROWN POINT = 926.81
2 - TOP NUT HYD. SO. SIDE LOT 9 BLK 1, CROWN POINT = 941.19

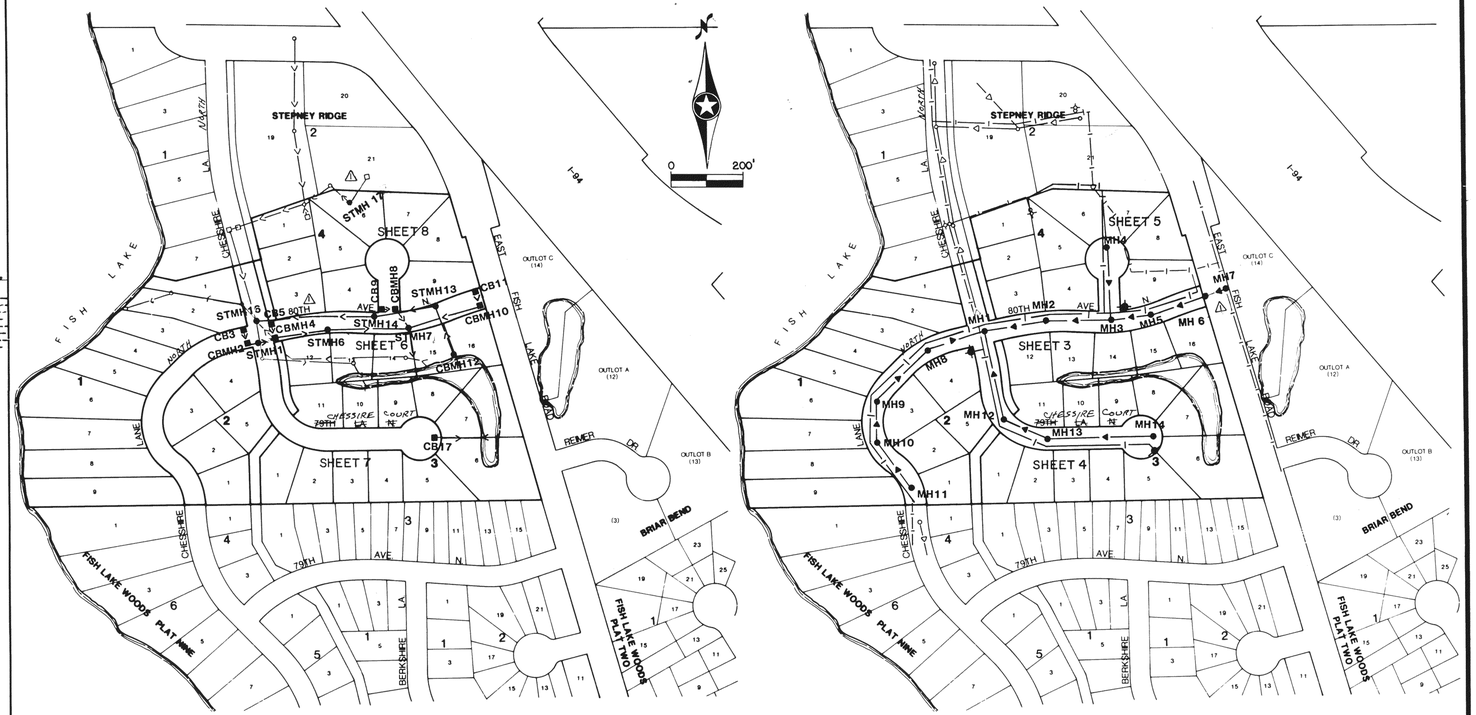
AS BUILT
MAR. 1988 H.H.

CONTRACTOR: S.J. LOUIS CONST. INC.

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE STATE OF MINNESOTA STATUTES SECTIONS 326.02 TO 326.16

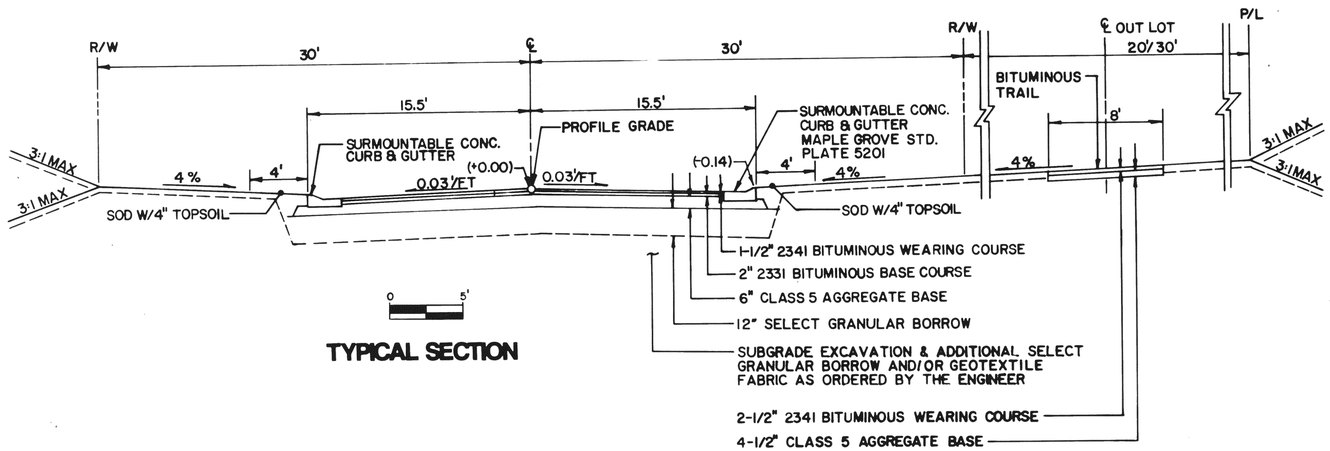
John C. Bartel DATE 8/3/87 MINN. REG. NO. 14960





STREET & STORM SEWER

SANITARY SEWER & WATERMAIN

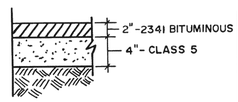


TYPICAL SECTION

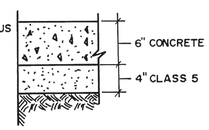
LEGEND

- EXISTING SANITARY SEWER
- PROPOSED SANITARY SEWER
- EXISTING STORM SEWER
- PROPOSED STORM SEWER
- EXISTING WATERMAIN
- PROPOSED WATERMAIN

BITUMINOUS



CONCRETE



DRIVEWAY RESTORATION DETAIL

CASTING ASSEMBLIES		
TYPE	NO. REQ'D	NEENAH NO.
CATCH BASIN	9	R-3246-G
STORM M.H.	7	R-1642-B
SANITARY M.H.	14	R-1642-B
CATCH BASIN	1	R-4342

As Built
MARCH 1988 H.H.

I hereby certify that this plan, specification or report was prepared by me or under my direct supervision and that I am a duly Registered Professional Engineer under the State of Minn. Statutes Sections 326.02 to 326.16.

John C. Bartel

Date 8/3/87 Minn. Reg. No. 14960

BRW
Barnett, Ringrose, Wiseland, Jarvis, Gardner
612-370-0700
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS: OMIT
OMIT STMH 15
ADD STMH 17 8-27-87

DESIGN BY: MRP
DRAWN BY: JAH, DM
SURVEY BK. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

CROWN POINTE

PROJECT INFORMATION & TYPICAL SECTION

S.A.P. NO.

SHEET 2 OF 8

DESIGN FILE No. 14-8722

502922

CHESSHIRE LANE

80TH AVENUE NORTH

BERKSHIRE LANE N.

EAST FISH LAKE ROAD

OUTLOT A

OUTLOT B

OUTLOT C

6"-1/16 BEND

6"-1/32 BEND

6"WV

6"W.V.

6"x6" CROSS

6"x6" TEE

4" INSTALL SALVAGED HYDRANT & VALVE

LOWER WATERMAIN W/ VERTICAL BENDS

10' DRAINAGE & UTILITY EASEMENT

EXISTING STORM SEWER TO BE SALVAGED

CONNECT TO EXISTING 6" D.I.P.

4"PVC SEWER SERVICE (TYP)
1"COPPER WATER SERVICE (TYP)
EXTEND 9' PAST LOT LINE

TYPICAL SERVICE WATER & SEWER SAME DITCH

SALVAGE HYDRANT & VALVE

INSTALL SALVAGED HYDRANT & VALVE

CONNECT TO EXISTING 6" D.I.P, REMOVE & REINSTALL 6"-1/4 BEND

7-23-87
4.8' FROM PAVEMENT
TO TOP NUT OF 6" W.V.

6'-1/4 BEND

10' MIN.

60' - 22'-24" STEEL CASING JACKED IN PLACE

10'-6"PVC, SDR 35 @ 0.40%
IE 917.50

FUTURE SERVICE TO BUILDING BLOCK DAY CARE CENTER

SALVAGE & REINSTALL VALVE

SALVAGE & REINSTALL HYDRANT & VALVE

ADJUST MANHOLE
EXISTING RIM 939.69
PROPOSED RIM 938.23

CONNECT TO EXISTING 6" DIP

LOWER ± 90 LF 6" DIP

SEE SHEET 4

NOTE:
INSTALL RISER SECTION ON SANITARY SERVICES TO LOTS 13 AND 14, BLOCK 3 TO GO OVER 42" RCP STORM SEWER MAINTAIN 12" CLEAR DISTANCE BETWEEN PIPES.

N.W.L. 914.50

MH1 MH2 MH3 MH4 MH5 MH6 MH7 MH8 MH9 MH10 MH11 MH12

0 50

EXISTING PROFILE

5.0%

S" DIP

± 90 L.F.

LOWER EXISTING W.M.

+3.63%

7.5' MIN.

PROPOSED ℄ GRADING GRADE

MH10

MH9

179'

+0.94% MH8

6" DIP CL 52

LOWER WATERMAIN W/ VERTICAL BENDS

MH1

℄ CHESSHIRE 927.35

TANGENT GRADE = -3.56

MH2

℄ BERKSHIRE 932.39

MH3

MH5

+3.44%

MH6

-30%

MH7

62' 24" STEEL CASING JACKED IN PLACE

PROPOSED 42" RCP

PROPOSED 15" RCP

6" DIP CL. 52

8" PVC SDR 35 @ 0.40%

8" PVC SDR 35 @ 1.27%

MH 11 (13+13 ℄)
RE 938.60 937.08
IE 919.00 917.09
DEPTH 19.60 19.99
19.99

MH 10 (11+48 ℄)
RE 933.91 933.78
IE 918.34 916.24
DEPTH 15.57 17.54
15.72

MH 9 (10+44 ℄)
R.E. 930.94 931.89
I.E. 917.92 917.67
DEPTH 13.02
13.42

MH 8 (8+35 ℄)
R.E. 928.45 928.57
I.E. 915.28 915.12
DEPTH 13.17
13.42

MH 1 (6+72.66 ℄)
R.E. 927.30 927.28
I.E. 914.63 914.22
DEPTH 12.67 12.98
12.98

MH 2 (4+95 ℄)
R.E. 931.27 931.42
I.E. 915.34 914.88
DEPTH 15.91 16.54

MH 3 (5 ℄)
R.E. 932.34 932.39
I.E. 916.06 915.64
DEPTH 16.28 16.75

MH 5 (2+10 ℄)
R.E. 929.54 929.63
I.E. 916.48 916.17
DEPTH 13.05 13.46

MH 6 (0+32 ℄)
R.E. 923.91 924.15
I.E. 917.19 916.87
DEPTH 6.72 7.28

MH 7 (-35 ℄)
RE 925.10
IE 917.46
DEPTH 7.46

935.25 935.56 937.56 936.76 939.16 937.57 937.11 935.78 934.23 933.98 932.36 931.11 930.19 929.66 929.69 929.12 926.65 926.18 927.70 927.55 927.35 927.52 928.25 929.55 930.15 931.17 932.29 930.83 932.77 932.14 930.92 930.34 929.24 927.52 925.79 924.42 924.15 924.33

940 935 930 925 920 915

106.43 PVT, MATCH EXISTING

+36.43 PVC +57 PVT +57 PVC +32 PVT +32.00 PVC +60 PVT +20 PVC

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

As Built
March 1988 H.H.

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Date 8/3/87 Minn. Reg. No. 19960

BRW
Bennett, Ringrose, Wolsfeld, Jarvis, Gardner
612-370-0700
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS
08-27-87, LOWER SAN. MH 9 - MH 11

DESIGN BY: MRP
DRAWN BY: LM, DM

SURVEY BK. NO.

S.A.P. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

CROWN POINTE

CHESSHIRE LANE/80TH AVENUE N.
SANITARY SEWER & WATERMAIN

SHEET 3 OF 8

DESIGN FILE No. 14-8722

SEE SHEET 3

FISH LAKE

80TH AVENUE NORTH

CHESSHIRE COURT

OUTLOT A

OUTLOT B

OUTLOT C

SERVICE REMOVAL NOTES

1) REMOVE 4" SERVICE TO WYE AND PLUG WYE

2) REMOVE 1" COPPER SERVICE TO CORPORATION AND TURN CORPORATION CLOSED

TIES
Ex. HYDRANT = 105.5'
MH 1 = 162'
WYE Loc. = 1491 L
IE 921.5

REMOVED 91' OF 4" & 1" SERVICE. CAPPED AND REINSTALLED CURB STOP

TIES
Ex. HYDRANT = 104.4'
EXISTING MH 4 = 79'
WYE Loc. = 0154 L
IE 920.2
100.06

TIES
Ex. HYDRANT = 162'
MH 1 = 78'
WYE Loc. = 2+85 L
IE 923.8

EXISTING MH 4

Ex. HYDRANT

109.06

REMOVE EXISTING SERVICE(1" WATER, 4" SAN.)

62.18

6"x6" CROSS

6" W.V.

REMOVE EXISTING SERVICE (1" WATER, 4" SAN)

DEPRESS WATERMAIN W/ VERTICAL BENDS

6" W.V.

PROPOSED 15" RCP

PROPOSED 42" RCP

REMOVE EXIST. MANHOLE AND ±52 L.F. 8" PVC CONSTRUCT MH 1 OVER EXIST. LINE

EXIST. STORM SEWER TO BE SALVAGED

MH 1

MH 2

MH 12

MH 13

MH 14

MH 8

PC 2+41.26

PT 3+41.07

PC 5+56.07

PT 5+94.63

10' DRAINAGE & UTILITY EASEMENT

10' MIN.

6"-1/8 BEND

6"-1/16 BEND

HYDRANT 6" W.V.

HYDRANT 6" W.V. 6"x6" TEE

4"PVC SEWER SERVICE (TYP)
1" COPPER WATER SERVICE (TYP)
EXTEND 9' PAST LOT LINE

TYPICAL SERVICE
WATER & SEWER SAME DITCH

0 50'

EXISTING PAVEMENT

+2.00%

+0.50%

+3.21%

-2.88%

-1.00%

PROPOSED ℄ GRADE

GRADING GRADE

REMOVE EXISTING MANHOLE

1.79

7.5' MIN

8" DIP CL 52 WM

EXISTING 6" DIP

EXISTING 8" PVC

PROPOSED 42" RCP

253'-8" PVC SDR 35 @ 0.40%

125'-8" PVC SDR35 @ 0.40%

295'-8" PVC SDR @ 35 0.40%

LOWER WATERMAIN W/VERTICAL BENDS

MH 1 (6+73, ℄)
RE 927.30
IE 914.63 .28
DEPTH 12.67
13.02

MH 12 (4+55, ℄)
RE 932.92 .67
IE 916.64 .52
DEPTH 16.28
17.27

MH 13 (5+85, ℄)
RE 931.87 .93
IE 916.14 .71
DEPTH 15.73 .32

MH 14 (8+80, ℄)
RE 926.66 .98
IE 917.32 .08
DEPTH 9.33
9.90

*100 PVC MATCH EXISTING PAVEMENT

50 PVC

*102.14 ℄ 80TH AVENUE N

20 PVC

20 PVT

90 PVC

90 PVT

50 PVC

50 PVC

*94.63 CENTER C.D.S.

0 1 2 3 4 5 6 7 8

935 930 925 920 915

As Built
March 1988 H.H.

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John C. Bartel

Date 3/3/87 Minn. Reg. No. 14960

BRW
Barnett, Ringrose, Widfeld, Jarvis, Gardner
612-370-0700
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS:
△ OMIT SERVICE 8-27-87

DESIGN BY: MRP
DRAWN BY: DYC, LMF

SURVEY BK. NO.

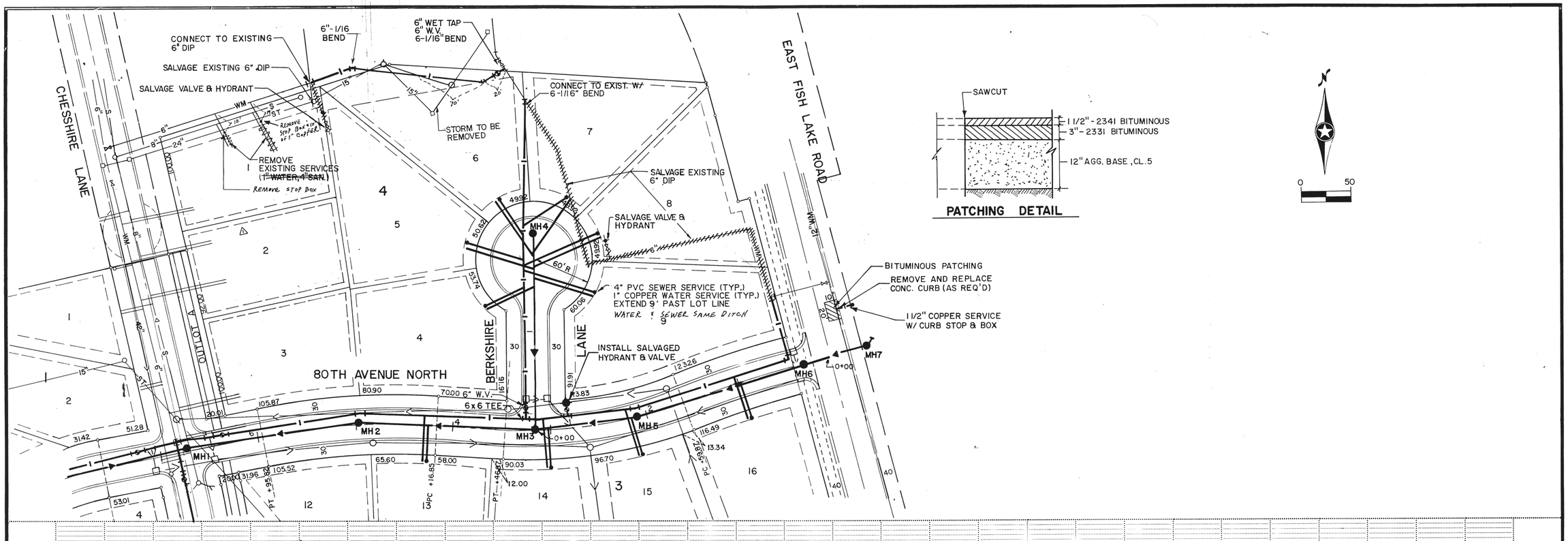
CITY OF MAPLE GROVE
PROJECT NO. 97-26

CROWN POINTE
CHESSIRE COURT
~~79TH LANE NORTH~~
SANITARY SEWER & WATERMAIN

S.A.P. NO.

SHEET 4 OF 8

DESIGN FILE No. 14-8722



As Built
March 1988 H.H.

I hereby certify that this plan, specification or report was prepared by me or under my direct supervision and that I am a duly Registered Professional Engineer under the State of Minn. Statutes Sections 326.02 to 326.16.

John C. Bratt

Date 8/3/87 Minn. Reg. No. 14960

BRW
Barnett, Ringrose, Wolfsfeld, Jarvis, Gardner
612 370 0700
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS:
△ OMIT SERVICE, ADJUST LOTLINE 8-27-87

DESIGN BY: MRP
DRAWN BY: LMF, DM
SURVEY BK. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

CROWN POINTE

BERKSHIRE LANE
SANITARY SEWER & WATERMAIN

S.A.P. NO.

SHEET 5 OF 8

DESIGN FILE No. 14-8722

CHESSHIRE LANE

80TH AVENUE NORTH

EAST FISH LAKE ROAD

BERKSHIRE LANE

SEE SHEET 7

SEE SHEET 8

SEE SHEET 8 FOR PROFILES

℄ CURVE DATA
R=142.48
Δ=98°00'00"
L=243.70
T=163.90

℄ CURVE DATA
R=270.08
Δ=33°00'00"
L=155.55
T=80.00

℄ CURVE DATA
R=151.44
Δ=52°41'36"
L=139.28
T=75.00

℄ CURVE DATA
R=683.62
Δ=15°00'00"
L=178.97
T=90.00

℄ CURVE DATA
R=428.52
Δ=25°00'01"
L=186.98
T=95.00

℄ CURVE DATA
R=1062.05
Δ=6°28'01"
L=119.87
T=60.00

10' DRAINAGE UTILITY EASEMENT

CONSTRUCT 8' BITUMINOUS TRAIL

CONSTRUCT 8' BITUMINOUS TRAIL

CONSTRUCT 8' BITUMINOUS TRAIL

30'-4" PVC PERF. PIPE

30'-4" PVC PERF. PIPE

30'-4" PVC PERF PIPE

SALVAGE RCP STORM SEWER PIPE & APRONS

SALVAGE EX. 42" CMP

OUTLOT A

OUTLOT B

OUTLOT C

CB3

CB5

CBMH2

CBMH 4

ST MH1

ST MH6

ST MH14

CB9

CBMH8

ST MH7

STMH13

CB11

CBMH10

CBMH12

42" RCP APRON W/ TRASH GUARD & SHEETING
136 C.Y. RANDOM RIPRAP CL III
36.0 S.Y. GEO-FABRIC FILTER

15" RCP APRON W/ TRASHGUARD
30 C.Y. RANDOM RIPRAP CL. III
9.0 S.Y. GEO-FABRIC FILTER

N.W.L. 914.50

BURRY & CARLSON PIT
EL 5 3½" HIGH AT CENTER OF C-D-S

REMOVE 60 L.F. EX. CONC. C & G

SAWCUT BIT. PAVEMENT

MATCH EXISTING PAVEMENT

EXISTING 8' BIT. PATH

(SEE DETAIL) RESTORE CONCRETE DRIVEWAY

REMOVE EXISTING BIT. PVMT. & CONCRETE CURB & GUTTER RECONSTRUCT BOULEVARDS EACH SIDE OF STREET

MATCH EXISTING PAVEMENT @ 15+06.43

(SEE DETAIL) RESTORE BIT. DRIVEWAY

ST MH1 (84" Ø)
R 1642-B
(6+98, 15.0' LT.)
RE ~~927.02~~ 926.98
N & E IE 916.~~50~~ .36
WIE 921.~~02~~ .50
DEPTH 10.62'

CBMH 2 (48" Ø)
R-3246-G
(7+09.16, 15.0' LT.)
RE 926.~~72~~ .63
EIE 921.~~72~~ .85
DEPTH 4.78'

CBMH 4 (66" Ø)
R-3246-G
(6+48.16, 15.75 LT.)
RE 926.~~78~~ .65
NIE 921.88
W & E IE 916.~~32~~ .20
DEPTH 10.45'

STMH 6 (66" Ø)
R-1642-B
(4+85, 20.0' LT.)
RE 931.~~51~~ .40
WIE 915.~~72~~ .62
DEPTH 15.78

STMH 7 (84" Ø)
R-1642-B
(2+63, 21.5' LT.)
RE 931.~~24~~ .32
NIE ~~926.00~~ 925.64
W & E IE 914.~~90~~ .94
SUMP IE 911.32
DEPTH 20.00

CBMH 8 (48" Ø)
R-3246-G
(2+99.58, 30.5' RT.)
RE 931.~~97~~ .85
WIE 927.~~55~~ .20
DEPTH 4.65'

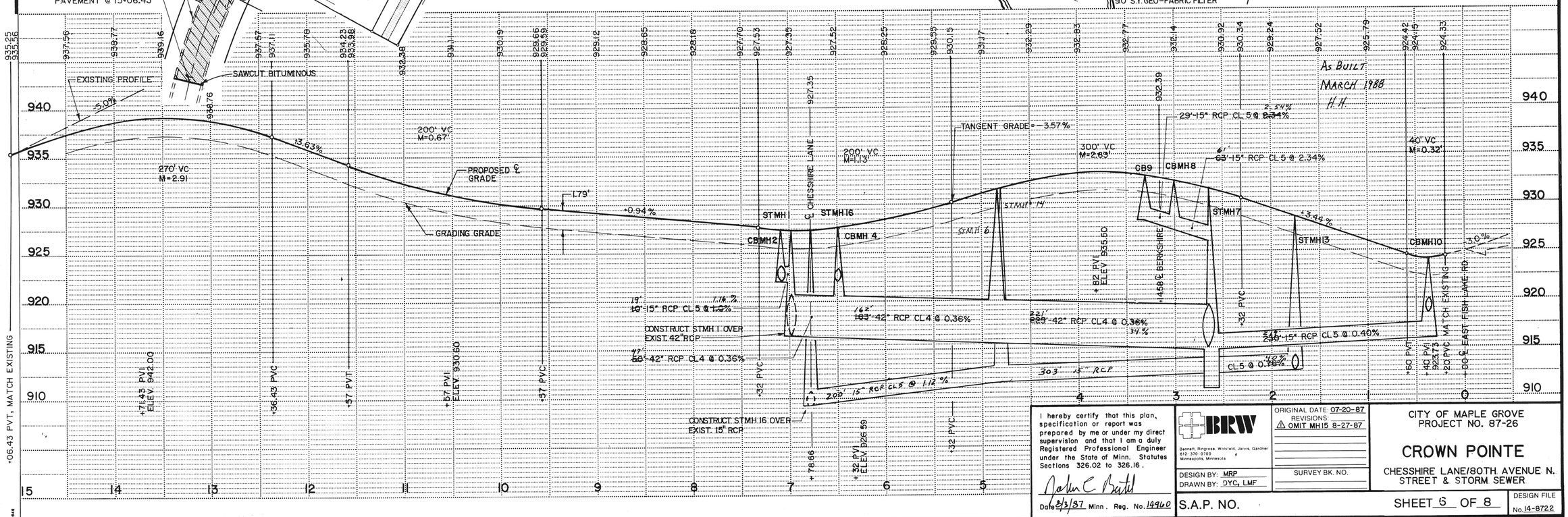
CB 9 (TYPE A)
R-3246-G
(3+29.08, 30.5' RT.)
RE 931.~~97~~ .85
IE 928.~~28~~ .10
DEPTH 3.75'

CBMH 10 (48" Ø)
R-3246-G
(0+37.00, 15.0' LT.)
RE 923.~~41~~ .84
NIE 918.~~41~~ .55
WIE 915.~~82~~ .95
DEPTH 7.89

STMH 13 (48" Ø)
R-1642-B
(1+75.00, 20.0' RT.)
RE 928.~~32~~ .43
SIE 912.~~74~~ .79
DEPTH 16.64'

STMH 14 (48" Ø)
R-1642-B
(3+45.08, 200' RT.)
RE ~~932.02~~ 931.21
WIE 911.~~48~~ .67
DEPTH 19.54'

STMH 16 (48" Ø)
R-1642-B
(6+78.66, 38.0' RT)
RE ~~927.11~~ 926.56
WIE ~~908.93~~ 908.69
DEPTH 17.87



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John C. Bartell

Date 8/3/87 Minn. Reg. No. 14960

BRW

Bennett, Ringrose, Wolsfeld, Jarvis, Gardner
612 370-0100
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS
⚠ OMIT MH15 8-27-87

DESIGN BY: MRP
DRAWN BY: DYC, LMF

SURVEY BK. NO.

S.A.P. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

CROWN POINTE

CHESSHIRE LANE/80TH AVENUE N.
STREET & STORM SEWER

SHEET 6 OF 8

DESIGN FILE No. 14-8722

SEE SHEET 6

CHESSHIRE LINE

80TH AVENUE NORTH

CONSTRUCT 6' BITUMINOUS TRAIL

OUTLOT A

REMOVE EXISTING BIT.

MATCH EXISTING PAVEMENT

SAWCUT BIT. PAVEMENT

STMH A
RE 926.34
IE 909.91
DEPTH 18.93

STMH B
RE 914.82
IE 899.01
DEPTH 15.81

STMH C
RE 895.34
IE 884.50
DEPTH 2.87

STMH 14

STMH 15

STMH 16

STMH 17

CBMH 4

CBMH 2

CB 5

CB 3

℄ CURVE DATA
R=661.27
Δ=8°38'53"
L=99.81
T=50.00

SALVAGE RCP STORM SEWER PIPE & APRONS

SALVAGE EX. 42" C.M.P.

10' DRAINAGE & UTILITY EASEMENT

OUTLOT B

CONSTRUCT 8' BITUMINOUS TRAIL

℄ CURVE DATA
R=147.33
Δ=85°00'00"
L=218.56
T=135.00

N.W.L. 914.50

15" RCP APRON W/ TRASHGUARD
3.0 C.Y. RANDOM RIPRAP CL. III
9.0 S.Y. GEO-FABRIC FILTER

I.E. 914.50

40'-4" PVC PERF. PIPE

BURRY & CARLSON PLACED
CL 5' 4½' HIGH AT
CENTER OF C-D-S

CHESSHIRE COURT

FISH LAKE

15" RCP APRON

31'-15" RCP @ 2.2 %

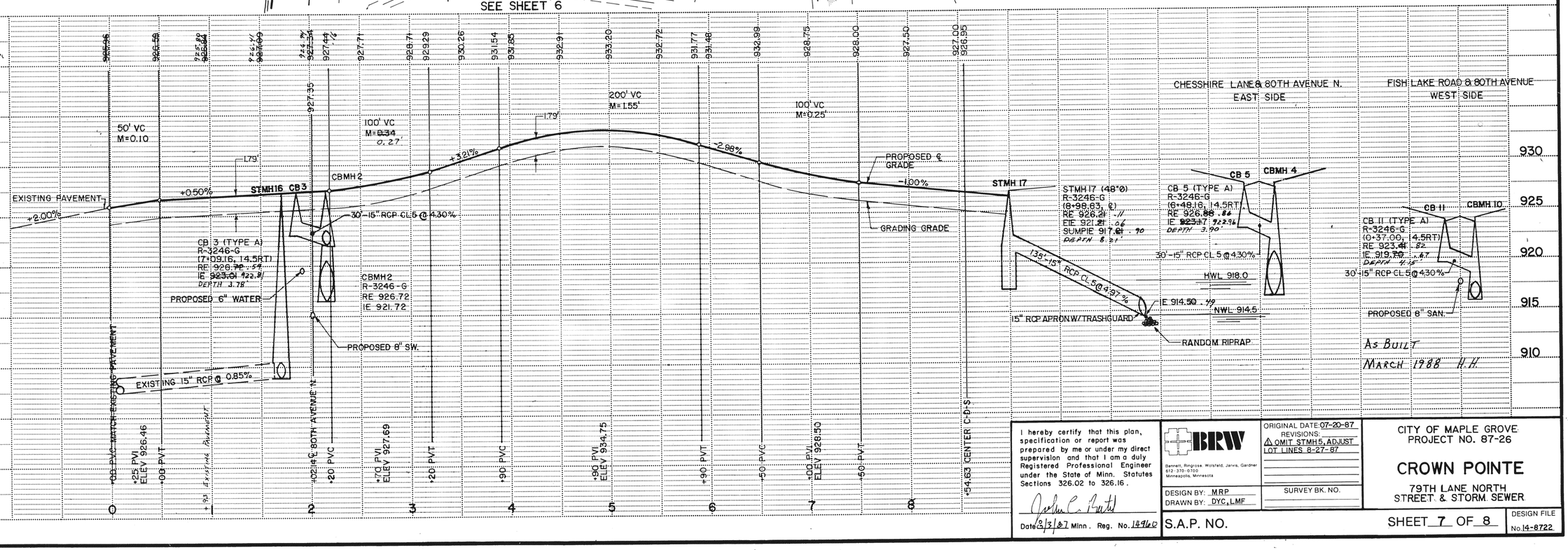
73'-15" RCP @ 1.8 %

148'-15" RCP @ 8.4 %

51'-15" RCP @ 2.6 %

0 50'

SEE SHEET 6



AS BUILT
MARCH 1988 H.H.

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Date 8/3/87 Minn. Reg. No. 14960

BRW
Bennett, Ringrose, Wolfsfeld, Jarvis, Gardner
612-370-0700
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS:
⚠ OMIT STMH 5, ADJUST LOT LINES 8-27-87

DESIGN BY: MRP
DRAWN BY: DYC, LMF
SURVEY BK. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

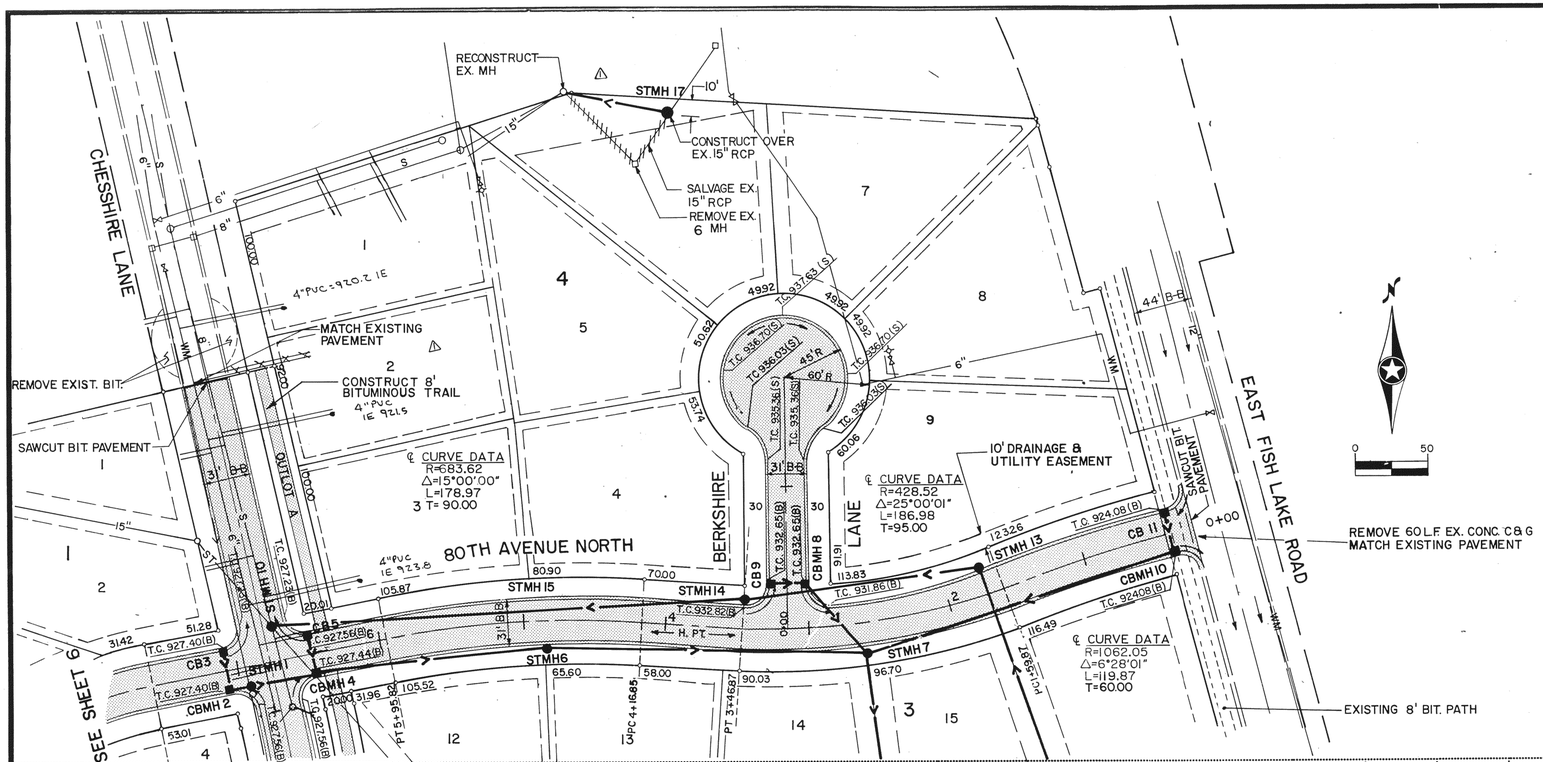
CROWN POINTE

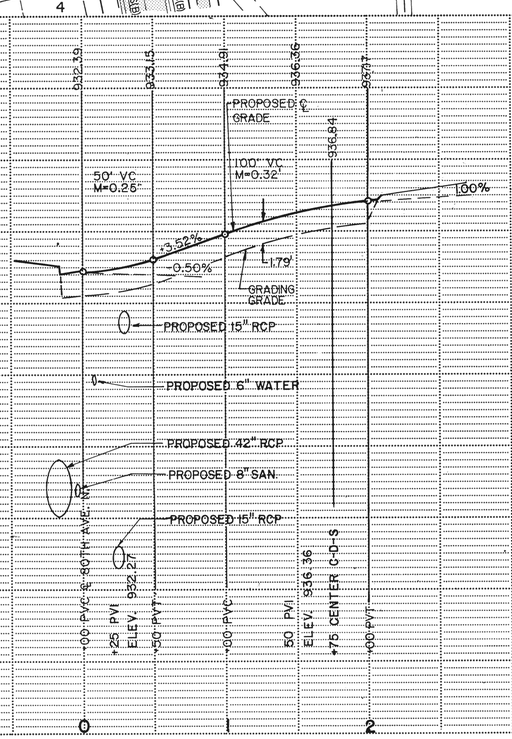
79TH LANE NORTH
STREET & STORM SEWER

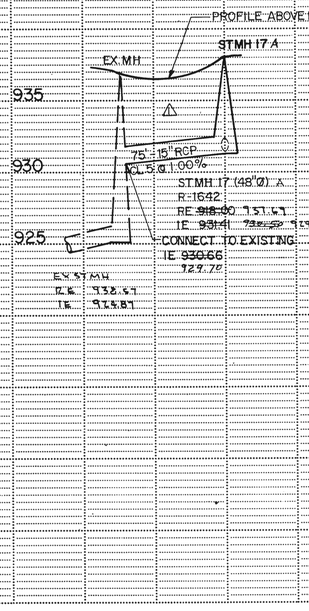
S.A.P. NO.

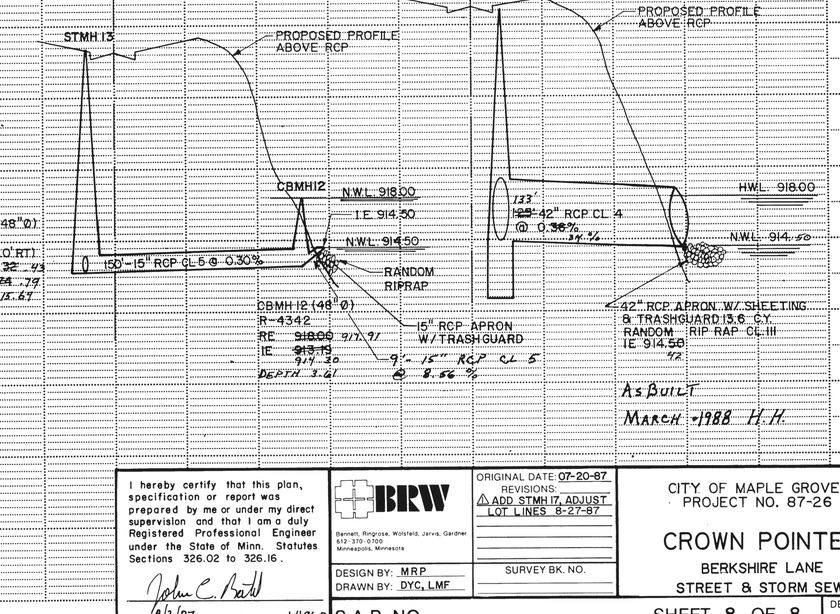
SHEET 7 OF 8

DESIGN FILE No. 14-8722









I hereby certify that this plan, specification or report was prepared by me or under my direct supervision and that I am a duly Registered Professional Engineer under the State of Minn. Statutes Sections 326.02 to 326.16.

John C. Bath

Date 8/3/87 Minn. Reg. No. 14960

BRW

Bennett, Ringrose, Wolsfeld, Jarvis, Gardner
612 370-0100
Minneapolis, Minnesota

ORIGINAL DATE: 07-20-87
REVISIONS:
⚠ ADD STMH 17, ADJUST LOT LINES 8-27-87

DESIGN BY: MRP
DRAWN BY: DYC, LMF
SURVEY BK. NO.

S.A.P. NO.

CITY OF MAPLE GROVE
PROJECT NO. 87-26

CROWN POINTE

BERKSHIRE LANE
STREET & STORM SEWER

SHEET 8 OF 8

DESIGN FILE No. 14-8722